

A Principled Bandwidth for Multitaper Coherence Estimation: Bias from Bessel Zeros, Variance from a Two-Tier Jackknife

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Abstract

Stage 4’s batch run confirmed a deterministic *upper* bound on the multitaper half-bandwidth W (equivalently the time-bandwidth product NW , via $K \approx 2NW$): the resolution criterion $2W \lesssim \Delta f_{\text{zero}} = c_{\text{min}}/(2r)$, where Δf_{zero} is the local spacing between the true coherence’s Bessel-function zero-crossings. What has been missing is a principled *lower* bound — how small NW can go before the coherence estimator’s own bias/variance behavior, not resolution, degrades picking. This document works out that lower bound following Haley & Anitescu’s (2017) mean-squared-error (MSE) framework for multitaper bandwidth selection, adapted in two ways specific to this project: (i) the bias term is derived in closed form from the known Bessel-zero structure of the theoretical coherence, rather than estimated by fitting a spline to noisy data, and (ii) the variance term is decomposed into two tiers — a taper-level term computable in closed form (compatible with FastMspec’s non-per-taper-materializing architecture) and a record-level term that separates genuine estimation noise from real day-to-day non-stationarity. Standalone reference, independent of the conversation history that produced it.

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1 Motivation

Ambient-noise interferometry estimates the coherence between two seismic stations, whose real part follows Aki’s (1957) spectral formulation,

$$\gamma(f) \approx J_0\left(\frac{2\pi fr}{c(f)}\right), \quad (1)$$

matching the convention of `coherence_barcode_design.tex` (r : interstation distance, $c(f)$: phase velocity). The FastMSPEC picker extracts a dispersion curve by tracking this function’s zero-crossings; the multitaper bandwidth W used to estimate $\gamma(f)$ from noisy data controls, as always in nonparametric spectral estimation, a bias–variance tradeoff.

The upper bound is already established. A bandwidth too wide smears adjacent zero-crossings together; the resolution criterion

$$2W \lesssim \Delta f_{\text{zero}}(f) = \frac{c(f)}{2r} \quad (2)$$

gives $NW_{\text{high}}(r) \approx Nc_{\text{min}}/(4r)$ for a window of N samples, where c_{min} is the *minimum* phase velocity over the analysis band (not the mean or the range’s upper bound — the constraint is tightest wherever $c(f)$ is smallest). This was confirmed directly against Round 1’s real data: quartile-4 pairs (780–1030 km) have $NW_{\text{high}} \approx 3.2$ –5.1, already below the project’s fixed operating point of $NW \approx 10.8$, a quantitative explanation for their observed near-zero convergence rate.

The lower bound is not. Today’s NW values (13–15 for MspecBestK, 80 for Mspec, FastMspec’s much smaller effective taper count) are hand-picked, not derived. This document addresses that gap.

2 The MSE Framework (Haley & Animescu, 2017)

Haley & Animescu [2] give a systematic method for multitaper bandwidth selection: minimize the estimated MSE of the log spectrum,

$$\hat{\Lambda}(f; N, W) = \frac{W^4}{36} \left[\frac{S''(f)}{\bar{S}(f)} \right]^2 + \hat{\sigma}_J^2(f), \quad (3)$$

summed over a frequency grid, where the first term is squared bias (from a second-order Taylor expansion of the true spectrum, Section 3) and $\hat{\sigma}_J^2(f)$ is a jackknife variance estimate (Section 4). The optimal NW minimizes the sum. Two features of their method do not transfer directly to this project and are addressed below: (i) their bias term is estimated from a spline fit to the *observed*, noisy multitaper spectrum itself — appropriate when nothing more is known about the true spectrum, but this project has a much stronger prior, the known Bessel-zero structure of Equation 1; and (ii) their variance theory is for a single-record auto-spectrum, whereas this project’s coherence estimate is a cross-spectral quantity additionally stacked over many independent day/window records (`coh_num`, often hundreds to ~ 1600) — a second averaging layer their paper does not address.

3 Bias: A Closed Form from the Bessel-Zero Structure

Haley & Animescu’s local bias term (their Eq. 10, here Eq. 3’s bias piece) is

$$B_W(f) \approx \frac{W^2}{6} S''(f), \quad (4)$$

the residual left by a symmetric smoothing kernel acting on the true spectrum’s local curvature. We show this applies *exactly* at a true zero-crossing f_n of Equation 1, not merely by analogy, and that $\gamma''(f_n)$ is computable in closed form.

3.1 The bias formula holds exactly at f_n

Write the true coherence near its zero at f_n as $\gamma(f) \approx \gamma'(f_n)(f - f_n) + \frac{1}{2}\gamma''(f_n)(f - f_n)^2$. Smoothing with a symmetric kernel K_W (the multitaper spectral window, Eq. 7 of the Haley–Anitescu supplement):

$$\bar{\gamma}(f_n) = \int K_W(f_n - \nu)\gamma(\nu) d\nu \approx \gamma'(f_n) \underbrace{\int K_W(u)u du}_{=0} + \frac{1}{2}\gamma''(f_n) \int K_W(u)u^2 du. \quad (5)$$

The first-derivative (slope) term vanishes identically for any symmetric kernel — an odd integrand over a symmetric window. What remains is exactly Eq. 4, evaluated at f_n : the smoothed estimate does not sit at zero, it sits at a spurious offset set purely by local curvature. This spurious residual is what a picker can mistake for a shifted or missing crossing.

3.2 Closed-form $\gamma''(f_n)$

With $\gamma(f) = J_0(x(f))$, $x(f) = 2\pi fr/c(f)$, and the Bessel recurrence $J_1'(x) = J_0(x) - J_1(x)/x$: at a zero x_n of J_0 (where $J_0(x_n) = 0$),

$$\gamma''(f_n) = J_1(x_n) \left[\frac{(x'(f_n))^2}{x_n} - x''(f_n) \right]. \quad (6)$$

Every term is computable without new data: x_n is the n -th zero of J_0 (exact, `scipy.special.jn_zeros`); $J_1(x_n)$ is a standard special-function evaluation, never zero at x_n by Bessel zero-interlacing; $x'(f_n)$, $x''(f_n)$ are derivatives of $2\pi fr/c(f)$, using the exact station distance r and $c(f)$ ’s local derivatives from the reference or picked curve already used for $c_{\min}$ in the Round 2 subset-selection script (`c_ref(f) + best_delta_km_s`).

Assumption, stated explicitly: this requires $c(f)$ to be locally well-behaved enough near each f_n for its derivatives to be meaningfully estimated from the picked curve — weaker than a globally constant c (already rejected for the $c_{\min}$ correction), but not guaranteed, especially in a rapidly-dispersive band. Worth checking empirically, not assumed.

3.3 Payoff

This replaces Haley & Anitescu’s self-referential spline-on-noisy-data bias estimate with a closed-form bias term derived from known physics, evaluated exactly where it matters (the zero-crossings themselves). No fitting step, no dependence on the very estimate whose quality it’s meant to characterize.

4 Variance: A Two-Tier Jackknife

4.1 Taper level (compatible with FastMspec’s architecture)

The delete-one jackknife variance of the log spectrum over K tapers (Haley–Anitescu supplement Eq. 23) needs each individual per-taper spectrum — straightforward for classical Mspec (which

already materializes per-taper spectra, chunked over K), but in tension with FastMspec’s deliberately non-per-taper-materializing architecture (Section 5). The supplement’s Eq. 24 resolves this: the *expected value* of the jackknife variance is a closed form in K alone (via the trigamma function ψ'),

$$E\{\hat{\sigma}_J^2(f)\} = \frac{(K-1)^3}{K \binom{K-1}{2}} \left[\frac{2}{(K-2)^2} + \frac{1}{2} \left(\psi' \left(\frac{K-1}{2} \right) - \psi' \left(\frac{K-2}{2} \right) \right) \right], \quad (7)$$

requiring only K (or FastMspec’s effective transition-region taper count, denoted r_{trans} to avoid collision with the distance symbol r above), not per-taper recomputation.

Update (2026-09-03), a real correction found by testing Eq. 7 against a literal Eq. 23 jackknife computed from real per-taper cross-spectra (SKRH-BAND pair, Mspec’s classical materialized-per-taper approach): Eq. 7 is not distribution-free, and it does not transfer cleanly to cross-spectra. Its derivation assumes each per-taper quantity is $\chi^2(2)$ -distributed — true for an auto-spectrum (the squared magnitude of one complex Gaussian) but not for a cross-spectrum per-taper term (the product of *two* correlated complex Gaussians), whose distribution near a coherence null is qualitatively different — the same $\sim 1/K$ near-null instability documented in Walden (2000). Binning the literal jackknife variance by per-taper magnitude (a proxy for proximity to a null) on real data shows the lowest-magnitude quartile has jackknife variance roughly 20–30 $\times$ the middle quartiles, and the literal-vs-closed-form ratio grows with K (1.9 $\times$ at $K = 5$, 8.3 $\times$ at $K = 33$) rather than converging — Eq. 7 systematically underestimates variance exactly where the picker’s crossing decisions are made.

Revised recommendation: use Eq. 7 away from candidate zero-crossings, where its assumptions are closer to valid and FastMspec’s memory-decoupling argument stays intact, but fall back to a literal per-crossing jackknife (or a genuinely cross-spectral closed form — Walden (2000)’s Wishart-based multivariate treatment [5] is the right place to check, not Haley & Anitescu’s univariate theory) in the neighborhood of each candidate crossing specifically. This is a bounded cost (a handful of crossings per pair, not the whole band), not a full retreat from Section 5’s argument — but it does mean Eq. 7 alone is not sufficient at the frequencies that matter most, and should not be presented as though it were.

4.2 Record level (a real complication, not glossed over)

Haley & Anitescu’s theory is single-record. This project stacks `coh_num` independent day/window traces on top of the K -taper average — a second layer their paper never addresses. Following the precedent in Park & Levin (2016) [4], who jackknife over *events* in a bin-average (not tapers) for exactly this multi-record situation (“a bin-average with 10 events leads to 10 different RF estimates,” via the delete-one method of Efron 1982), decompose the per-trace estimates $X_i(f)$ (log- or Fisher-transformed) as

$$X_i(f) = \mu(f) + \delta_i(f) + \varepsilon_i(f), \quad (8)$$

where δ_i is genuine day-to-day physical variation (non-stationarity: storms, source shifts) and ε_i is that day’s own K -taper estimation noise. Then

$$\widehat{\text{Var}}(\delta; f) \approx \text{SampleVar}_i(X_i(f)) - E\{\hat{\sigma}_J^2(f)\}, \quad (9)$$

using Eq. 7 as the estimation-noise baseline (cheap, FastMspec-compatible) and the plain sample variance across traces as the empirical total (Haley & Anitescu’s jackknife motivation, statistical efficiency for a *small* number of independent samples, applies most naturally at the K -taper tier; at `coh_num` scale, already in the hundreds, plain sample variance is simpler and no less appropriate).

This is a diagnostic, not just a variance estimate. Computing Equation 9 as a function of frequency identifies where the coherence estimate is unstable for a *physical* reason (real non-stationarity) rather than an estimator-tunable one (insufficient tapering). At frequencies where $\widehat{\text{Var}}(\delta; f)$ dominates, no choice of NW will stabilize picking — the MSE-minimization machinery of Section 6 is the wrong tool there, and that should be stated plainly in any picking-reliability discussion rather than attributed to a suboptimal bandwidth.

5 Why FastMspec’s Architecture Survives This Analysis

FastMspec (Karnik, Romberg & Davenport’s algorithm [3], translated in `fast_multitaper.py`) decomposes the multitaper average into an $O(N \log N)$ bulk sinc-convolution term plus a small correction over only the DPSS eigenvalue transition-region tapers, whose count r_{trans} stays bounded (11–24) across $NW = 5\text{--}1600$ while K grows linearly the whole way. This means (i) the memory argument for FastMspec over classical Mspec/MspecBestK holds regardless of where the bandwidth search in this document lands, and (ii) the taper-level variance term above (Eq. 7) never needs FastMspec’s fused computation broken open — K alone (or r_{trans} , if the transition-region-only variance is preferred) is sufficient.

6 Putting It Together

The full recipe, combining Sections 3–4 with the already-derived upper bound (Equation 2):

1. Compute $NW_{\text{high}}(r)$ per pair from its own c_{min} (Round 2 subset-selection script).
2. At each candidate NW in the sweep, compute the closed-form bias (Equation 6) at the theoretical zero-crossing frequencies, and the closed-form taper-level variance (Equation 7).
3. Minimize $\hat{A}(NW) = \text{bias}^2 + \text{variance}$, restricted to $NW \leq NW_{\text{high}}(r)$ (the resolution bound is a hard ceiling on catastrophic failure, not part of the smooth tradeoff; the true optimum can sit well inside it, as in Haley & Anitescu’s own Lorentzian example, $NW = 14.5$ against a much larger ceiling).
4. Separately compute the excess-variance diagnostic (Equation 9) to flag frequencies where bandwidth tuning cannot help at all.

Round 2’s NW sweep (60-pair stratified subset, 5 fractions of $NW_{\text{high}}(r)$ each) is the mechanism for computing this empirically once real per-fraction convergence/quality data exists — this document gives the theory the sweep is meant to test, not a substitute for running it.

7 Open Questions

- Whether $c(f)$ ’s local derivatives, needed for Equation 6, are reliably estimable from the picked/reference curve in rapidly-dispersive bands, or need their own smoothing/regularization.
- Whether δ_i and ε_i (Section 4) are actually independent, as assumed in Equation 9 — untested against real data.
- Whether the closed-form bias (Section 3) should use the fundamental-mode Bessel structure alone, or needs correction for multi-mode contamination, matching the same caveat already carried by the Love-wave Bessel-model simplification in `coherence_barcode_design.tex`.
- *Tested, not fully resolved:* the extent of the near-null breakdown documented in Section 4’s update – confirmed present and large on one real pair (SKRH–BAND), not yet checked across the dataset’s real distance/coherence range, nor against a genuinely cross-spectral closed-form alternative.

References

- [1] Aki, K. (1957). Space and time spectra of stationary stochastic waves, with special reference to microtremors. *Bulletin of the Earthquake Research Institute*, 35, 415–456.
- [2] Haley, C. L., & Anitescu, M. (2017). Optimal bandwidth for multitaper spectrum estimation. *IEEE Signal Processing Letters*, 24(11), 1696–1700. <https://doi.org/10.1109/LSP.2017.2719943>
- [3] Karnik, S., Romberg, J., & Davenport, M. A. (2022). Multitaper spectral estimation revisited: fast algorithms and generalized approaches. Preprint. (FastMSPEC project reference: `Multitaper_Revisited_Karnik.pdf`.)
- [4] Park, J., & Levin, V. (2016). Statistics of multiple-taper receiver functions. *Geophysical Journal International*, 207(1), 512–527. <https://doi.org/10.1093/gji/ggw291>
- [5] Walden, A. T. (2000). A unified view of multitaper multivariate spectral estimation. *Biometrika*, 87(4), 767–788. <https://doi.org/10.1093/biomet/87.4.767>
- [6] Ekström, G., Abers, G. A., & Webb, S. C. (2009). Determination of surface-wave phase velocities across USArray from noise and Aki’s spectral formulation. *Geophysical Research Letters*, 36(18), L18301. <https://doi.org/10.1029/2009GL039131>